

LLM Engineering Savol - Javob.

① Vocab ^{Level} ~~size~~ ^{BAS 30} nima anglatadi

vocab-size tokenlar lugatidagi jamo tokenlar sonini bildiradi misol u-n 8000 8000 ta token modeling embedding na 1M head qo'llanmo shu qiziqqa bog'langan

② block size nima?

Model bir vaqtning o'zida qancha data oladi, maksimal tokenlar soni qancha misol u-n 256 token bo'lsa u-n kamligiga qarab hisoblash na VRAM ni ko'rib chiqish kerak

③ n-emb nima?

Modelga kiritilgan Transformer block qoshadi. Bu rep-ni doqircha bo'lish chiqarib beradi na muvaffaqiyatli patternlarni o'rganish imkonini beradi

④ n-layer oshirilsa nima bo'ladi

har bir tokenning embedding / vector alchamini belgilaydi. 384 yoki har bir token 384 dimensional representation sifatida ishlatiladi

⑤ Attention heads nima?

Sequence ichidagi tokenlar orasidagi bog'ni mulohaza o'tirish sub-space orqali tuzilgan parallel mexanizm

Nimaga n-emb n-head ga bo'linishi kerak Emb-ni dimension headlar orasida teng taqsimlanadi
$$\text{head dim} = \frac{n\text{-emb}}{n\text{-head}}$$

⑥ dropout = 0.1 vasifosi nima?

Dropout training vaqtida o'qitishni
taxminan 20% qismini tasodifiy o'chirib
modelning training dataga kaddan
tasviri moslashishni kamaytiradi.

⑦ FFN nima va ifani bajaradi?

Har bir token representationni non-linear
trans-n orqali qayta shakllantirish va natijada
muvozanatli featurelarni o'rganish imkonini
beradi.

⑧ tie weights: true nima?

Input embedding matrix va output /H
head weight matrix bitta weightni
qo'llashishni anglatadi.
Bunda $1000 \times 304 = 3,04,000$ parameter alshada
weight sifatida saqlanmaydi.

⑨ Random-init, loss check nima?

Trainingdan oldin random-initalised
modelning loss sigmatini tekshiriladi. Bu
model tokenlarni va loss populyatsiya
to'g'ri shakllanishini tekshirish uchun base
line hisoblanadi.

⑩ Greedy Decoding nima?

Har bir qadamda eng
katta probabilitetga ega token
tanlanadi $\text{next_token} = \text{argmax}(\text{logits})$

Medium Level

① Causal^{self} va Global Attention farqi

Causal Attention shenni faqat oydan
oldingi token larga qarabgina maybur

Global Attention esa future token'ni
ham ko'rishni mumkin.

② Model size va Token budget nega farq

Modelda juda katta modelno juda kichik
datasetda faqat qilish compute va server
niqat nazaridan samaralar qilishi uch.

③ Nega weightlar std=0.2 ga initialize qilinadi?

Kichik random initializationdan tashqari
Symmetry breaking beradi va gradientlarning
ishlashi uchun boshlang'ich signal yaratadi.

④ Pre Norm nima uchun foydali

By gradient slowing barga o'rnatilgan
va chunki transformatsiya training
qilishni osonlashtiradi. Pre Normota
layer norm oson sublayerdan oldin
qo'yiladi.

⑤ Residual Connection nima beradi?

Resid con-n gradient uchun yo'roqdan yo'ri
yol yaratadi. By deep network da gradient
reanishonni miqnosini karyayotib
optimizatsionni osonlashtiradi.

⑥ Causality Test nima tekshoradi?

Modelning T pozitsiyadagi output
belgilanadigan $t+1, t+2, \dots$ tokenlarga bo'yliq
emasligini tekshoradi

⑦ FFN nega har position un
niqatigal shunday?

FFN tokenlar orasida communication
qilmaydi $FFN(x_i) = FFN(x_j) = FFN(x_k)$
har biri alohida hisoblanadi.

⑧ One-batch overfit testi nima un?

Model buda kichik batchda
o'ylaydi, biro loss ga cha'qana old
shind tekshoradi, bu model architecture
optimizer, loss va training lossdagi
xatolarni aniqlash un shunday.

⑨ $idx_cond = idx \cdot T_i$, $-block_size$ nima uchun?

Generation reagentda model faqat
o'zgi block size tokenni context size
ida boradi.

⑩ Mega configuration YAML faylda?

Hyperparameterlarni koddan o'z
ish experimentlarni boshqarish
va reproducibilityni yaxshi layda

Advanced Line

① **Mega production** MHA o'rnida GQA
 GQA - bu nishda query head uchungu kanva
 key value head headan foydalanadi
 Bu attention so'zini so'zlagan
 holida kv cache haymini kamaytira

② **num-parameters()** juda ko'patsini qanday
 hisoblaydi?

Tog'ri hisoblashda bor xil storage/param.
 qila shki ma'na sanalmaydi. Tada
 Embedding va LM head bitta niqos
 ni chiqarib qoladi

③ **Learned Positional Embedding** va **RoPE** farqi?
 PBU har bir position u-n tashkil etadi
 dan vector saqlaydi. RoPE esa
 Query va key vectorlarini positionga
 bog'liq rotatsion argali tashkil etadi

④ **SwiGLU va GELU FFN** farqi
 Oddiy Gelu $\rightarrow$ linear $\rightarrow$ gelu $\rightarrow$ linear
 xolatda. SwiGLU esa gated structure
 tashkil etadi va qo'shimcha projection
 argali activationni bo'lisha radi

⑤ **Mega frontier model** trainingda
 dropout ko'pincha ishlatilmaydi.
 Juda katta Dataset na juda katta
 token budget modelni oddiy ravishda
 regularize qiladi. Dropout esa training
 signalini samaradorlikni mu-

⑤ Mega logits (view(-1, logits, seq(-1)))
 Transformer sequence, vocab, logits, softmax
 Cross entropy loss, vocab, softmax
 batch size, sequence dimension

⑥ Initial loss 15.0, softmax, vocab, softmax
 baseline, softmax, softmax, softmax
 labels, loss calculation, softmax, softmax
 pipeline, softmax, softmax, softmax

⑦ Mega logits[1, -1, 1] softmax?

Generators, softmax, softmax, softmax
 softmax, softmax, softmax, softmax
 softmax, softmax, softmax, softmax

⑧ Self Attention, quadratic complexity, softmax

Standard self-attention, softmax, softmax
 softmax, softmax, softmax, softmax
 softmax, softmax, softmax, softmax

⑨ "Bigger model automatically better"
 softmax, softmax, softmax, softmax

Model Quality, softmax, softmax
 softmax, softmax, softmax, softmax
 softmax, softmax, softmax, softmax